

KITAMIESASHI, Japan

WMO: 474020

Lat: 44.9400N

Lon: 142.5850E

Elev: 8

StdP: 101.22

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-11.9	-17.4	0.8	-12.6	-15.8	0.9	-11.2	13.2	-1.4	11.6	-1.9	2.8	240	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	27.2	20.8	25.3	20.0	23.6	19.4	22.4	25.7	21.3	24.0	20.3	22.7	5.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	15.9	24.6	20.2	14.9	23.3	19.3	14.0	22.2	66.1	25.7	62.2	24.1	58.6	22.7	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.7	9.2	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-15.9	31.0	1.8	1.8	-17.3	32.2	-18.3	33.3	-19.4	34.3	-20.7	35.6		
			-16.4	23.8	1.8	1.3	-17.7	24.8	-18.7	25.6	-19.7	26.3	-21.0	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-5.9	-5.9	-1.6	4.4	9.3	12.6	16.7	19.4	16.7	10.3	2.9	-3.2	(6)
(7)		DBStd	9.46	3.09	3.63	3.64	3.84	4.63	3.91	3.64	3.11	2.98	3.38	4.19	3.18	(7)
(8)		HDD10.0	2218	492	444	359	173	71	14	0	0	0	38	216	410	(8)
(9)		HDD18.3	4468	750	678	618	417	282	177	77	24	67	250	462	668	(9)
(10)		CDD10.0	901	0	0	0	7	49	94	209	292	202	46	3	0	(10)
(11)		CDD18.3	109	0	0	0	0	2	6	27	57	18	0	0	0	(11)
(12)		CDH23.3	436	0	0	0	0	19	32	116	220	48	1	0	0	(12)
(13)		CDH26.7	77	0	0	0	0	3	5	23	42	4	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.5	3.3	3.7	3.7	3.5	2.9	2.6	2.9	3.4	3.6	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	1098	69	54	59	57	69	72	121	130	143	111	115	98	(15)
(16)		PrecMax	1489	134	117	114	125	139	196	296	329	369	167	223	139	(16)
(17)		PrecMin	693	32	24	6	4	21	20	16	35	49	44	63	57	(17)
(18)		PrecStd	175	25	23	25	27	30	35	70	76	72	39	42	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	6.2	10.1	19.9	26.2	26.9	29.3	30.1	26.9	21.0	15.7	7.5	(19)
(20)			MCWB	0.6	2.8	5.0	11.0	14.6	17.1	22.6	23.0	20.1	15.3	10.8	5.1	(20)
(21)		2%	DB	0.8	3.4	7.3	15.7	21.8	23.4	26.2	27.7	24.6	18.6	13.0	4.7	(21)
(22)			MCWB	-0.6	0.8	3.6	8.8	13.2	16.0	21.0	21.8	19.2	13.6	9.1	2.7	(22)
(23)		5%	DB	-0.1	1.3	5.3	12.7	18.9	20.5	24.2	25.8	22.9	17.0	11.0	2.9	(23)
(24)			MCWB	-1.3	-0.7	2.3	7.1	12.1	15.2	20.3	21.0	18.3	13.0	8.0	1.3	(24)
(25)		10%	DB	-1.1	-0.5	3.6	10.3	16.2	18.3	22.1	23.9	21.4	15.6	9.0	1.5	(25)
(26)			MCWB	-2.3	-2.0	1.0	5.9	10.9	14.6	19.3	20.5	17.5	12.0	6.6	0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.0	3.1	5.9	11.7	15.9	18.9	23.6	24.1	21.8	16.4	11.9	5.4	(27)
(28)			MCDWB	2.2	5.7	9.3	18.2	22.7	24.6	27.5	29.1	25.5	19.6	15.0	7.2	(28)
(29)		2%	WB	-0.2	1.1	4.0	9.4	14.1	17.1	22.0	22.8	20.5	14.8	9.7	3.0	(29)
(30)			MCDWB	0.6	3.1	6.9	14.8	20.2	21.6	25.4	26.3	23.2	17.4	12.3	4.4	(30)
(31)		5%	WB	-1.3	-0.6	2.5	7.7	12.7	15.9	20.7	21.9	19.2	13.7	8.2	1.5	(31)
(32)			MCDWB	-0.3	1.0	4.9	11.8	18.2	19.4	23.5	24.7	21.8	16.3	10.4	2.7	(32)
(33)		10%	WB	-2.3	-1.8	1.2	6.3	11.3	14.8	19.4	20.9	18.1	12.5	6.8	0.2	(33)
(34)			MCDWB	-1.2	-0.5	3.2	10.0	15.5	17.8	21.9	23.1	20.7	15.1	9.0	1.4	(34)
(35)	Mean Daily Temperature Range	MDBR	5.2	6.3	6.1	6.8	6.9	5.7	4.9	5.2	7.1	7.4	5.7	5.0	(35)	
(36)		5% DB	MCDBR	5.7	8.0	8.4	11.1	12.0	10.3	9.0	8.3	8.7	9.1	8.4	6.3	(36)
(37)			MCWBR	5.2	6.5	5.9	6.3	5.9	5.3	4.8	4.2	4.9	5.7	6.1	5.5	(37)
(38)		5% WB	MCDBR	5.4	7.4	8.0	10.3	11.2	8.7	8.0	7.1	7.3	7.9	7.7	6.1	(38)
(39)			MCWBR	5.2	6.3	6.0	6.4	5.8	5.0	4.8	4.1	4.7	5.6	6.0	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.336	0.381	0.443	0.508	0.524	0.504	0.481	0.453	0.404	0.406	0.387	0.356	(40)	
(41)		taud	2.032	1.861	1.807	1.820	1.880	2.030	2.162	2.259	2.338	2.168	2.114	2.035	(41)	
(42)		Ebn at Noon	728	756	765	755	760	778	792	801	813	737	667	652	(42)	
(43)		Edh at Noon	105	152	185	199	194	167	145	127	108	111	98	96	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.21	2.12	3.32	4.42	4.87	4.82	4.50	4.18	3.63	2.47	1.31	0.97	(44)	
(45)		RadStd	0.13	0.28	0.30	0.38	0.42	0.53	0.58	0.30	0.32	0.16	0.14	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.16	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page